Yuanming Lu

Mathematical Biology | **Theoretical & Spatial Ecology** | **Biological Invasions**

Gainesville, FL | (631) 496-6954 | yuanming.lu@ufl.edu | yuanming110.github.io

Quantitative Biology Profile

Theoretical ecologist using mathematical and computational models to investigate dispersal, species interactions, biological invasions, and disease transmission in heterogeneous environments. Research compares individual-based, metapopulation, and reaction–diffusion models to connect local mechanisms with population and landscape dynamics. USGS- and USDA-supported collaborations link ecological theory with environmental data, invasive-species management, and ecosystem restoration. Experienced in computational research, undergraduate biology instruction, and student mentoring.

Education

University of Florida Aug. 2026
Ph.D. in Zoology *Gainesville, FL*

- Advisors: Dr. Robert D. Holt and Dr. Jeremy W. Lichstein.
- Dissertation: *Why Individuals Matter: Spatially Explicit Models of Plant Invasion, Dispersal, and Biocontrol.*

University of Miami May 2021
M.S. in Biology *Coral Gables, FL*

- Advisor: Dr. Donald L. DeAngelis.
- Thesis: *Spatial Pattern Formation of Invasive Melaleuca Vegetation: Interspecies Interactions, Regime Shift, Biocontrol, and Possible Early Warning Signal.*

State University of New York at Stony Brook Dec. 2017
B.S. in Biology, Ecology and Evolution *Stony Brook, NY*

Academic Appointment

University of Florida, Department of Biology Oct. 2026–Present
Postdoctoral Associate *Gainesville and Fort Pierce, FL*

- Research collaborators: Dr. Robert D. Holt and Dr. Carey R. Minteer
- Conduct research on integrated management of waterhyacinth (*Pontederia crassipes*) under USDA award 58-6032-2-005, in collaboration with Drs. Holt and Minteer.

Research Experience and Externally Supported Collaborations

Holt Lab, University of Florida Aug. 2021–Aug. 2026
Doctoral Researcher *Gainesville, FL*

- Developed metapopulation, reaction–diffusion, and individual-based models to examine biological invasions, spatial competition, dispersal, and range expansion in heterogeneous landscapes.
- Compared deterministic and stochastic formulations to determine how individual-level processes and model structure affect predictions of coexistence and spatial spread.
- Designed simulation experiments, analyzed model outputs, and led manuscripts linking dispersal, local interactions, and environmental heterogeneity to population and landscape dynamics.

Collaborative Animal Disease Ecology Project Aug. 2026–Present
Researcher *Remote collaboration*

- Developing a spatial model with Jing Jiao to distinguish environmental connectivity and live-fish relocation as pathways of infectious salmon anaemia transmission among aquaculture sites.
- Integrating public farm, disease, and environmental data with movement scenarios to examine how distinct connectivity pathways shape regional disease transmission.

USGS Greater Everglades Priority Ecosystem Science Program 2021–2025
Funded Research Collaborator, University of Florida *Gainesville, FL*

- Developed a spatially explicit individual-based model of *Melaleuca quinquenervia* invasion incorporating demography, seed dispersal, tree legacies, and landscape heterogeneity.
- Evaluated invasion and management scenarios relevant to invasive-plant control and ecosystem restoration; collaborated with university and agency scientists on publications and presentations.

USDA Invasive Species and Biocontrol Program 2024–2025
Funded Research Collaborator, University of Florida *Gainesville, FL*

- Developed agent-based models to evaluate biological control and the effects of interacting biotic stressors on invasive-plant dynamics.
- Integrated field- and literature-derived parameters, designed simulation experiments, and contributed to interdisciplinary proposal development and project reporting.

DeAngelis Lab, University of Miami Jan. 2019–May 2021
Graduate Researcher *Coral Gables, FL*

- Developed spatially explicit models of invasive-plant spread, litter feedbacks, biological control, and mechanical treatment; designed simulation experiments and contributed to peer-reviewed publications.

Computational and Analytical Expertise

Ecological modeling: individual-based, metapopulation, spatially explicit, reaction–diffusion, deterministic, and stochastic models.
Data analysis: simulation experiment design, environmental and spatial data integration, model comparison, sensitivity analysis, ecological visualization, and management scenario evaluation.
Programming and workflows: Python, R, MATLAB, NetLogo, Git/GitHub, reproducible research, documented computational workflows, and high-performance computing (HiPerGator).

Peer-Reviewed Publications

Seven published articles; four first-author articles. Yuanming Lu is shown in bold.

1. **Lu Y**, Lichstein JW, Khare SB, Holt RD, Yurek S, DeAngelis DL. 2026. Slow and fast dispersers competing in heterogeneous landscapes: a comparative study of modeling perspectives. *Journal of Theoretical Biology* 626:112448. doi:10.1016/j.jtbi.2026.112448.
2. Zheng C, Magee L, Cayetano D, Rocha KD, **Lu Y**, Bohlman S, Anderson-Teixeira K, Lutz JA, Silva C, Johnson DJ. 2026. Discriminating temporal decline and death of a savanna oak using multi-year imaging spectroscopy. *Remote Sensing Applications: Society and Environment* 43:102117. doi:10.1016/j.rsase.2026.102117.
3. **Lu Y**, Xia J, Holt RD, DeAngelis DL. 2024. Modeling the effects of spatial distribution on dynamics of an invading *Melaleuca quinquenervia* population. *Forests* 15:1308. doi:10.3390/f15081308.
4. Magee LJ, LaManna JA, Wolf AT, Howe RW, **Lu Y**, Valle D, Johnson DJ. 2024. The unexpected influence of legacy conspecific density dependence. *Ecology Letters* 27:e14449.
5. **Lu Y**, Xia J, Magee LJ, DeAngelis DL. 2023. Seed dispersal and tree legacies influence spatial patterns of plant invasion dynamics. *Frontiers in Applied Mathematics and Statistics* 9:1086781.
6. **Lu Y**, DeAngelis DL, Xia J, Jiang J. 2022. Modeling the impact of invasive species litter on conditions affecting its spread and potential regime shift. *Ecological Modelling* 468:109962.
7. Fu Z, **Lu Y**, DeAngelis DL, Zhang J, Zhang B. 2021. Integrating mechanical treatment and biological control to improve field treatment efficiency on invasions. *Elementa: Science of the Anthropocene* 9:00181.

Manuscripts Submitted, Under Review, and in Preparation

1. **Lu Y** (first and corresponding author), Lichstein JW, Holt RD, DeAngelis DL. 2026. Initial invasion footprint shapes biocontrol outcome in a spatially explicit individual-based model of *Melaleuca* invasion and biocontrol. Submitted to *Individual-based Ecology*.
2. **Lu Y**, Xia J, Khare SB, Gillooly JF, Zhang W, DeAngelis DL, Holt RD. Climate-driven spatial competition: a metapopulation patch model assessing range expansion in species with different dispersal rates. Manuscript under review.
3. **Lu Y**, Jiao J. Modeling environmental connectivity and live-fish relocation in the spatial transmission of infectious salmon anaemia. Manuscript in preparation.

Selected Presentations and Workshops

1. **Lu Y**, Lichstein JW, DeAngelis DL, Holt RD. 2026. Can biocontrol spread reduce the need for repeated releases to suppress *Melaleuca* invasion? *Ecological Society of America Annual Meeting*. Contributed talk.
2. **Lu Y**. 2025. ABM development in MATLAB and Python: examples from ecology. *ABM and NetLogo Workshop*. Contributed talk.
3. **Lu Y**, Lichstein JW, DeAngelis DL, Holt RD. 2025. When myrtle rust joins the fight: modeling biocontrol synergies against invasive *Melaleuca*. *Ecological Society of America Annual Meeting*. Contributed talk.
4. **Lu Y**, Xia J, DeAngelis DL, Holt RD. 2024. Hybrid approaches to invasive species management: agent-based modeling and image classification. *Ecological Society of America Annual Meeting*. Contributed talk.
5. **Lu Y**, Xia J, Magee LJ, DeAngelis DL, Holt RD. 2023. Unveiling the battlefront: *Melaleuca* invasion dynamics in the Everglades. *Greater Everglades Ecosystem Restoration Conference*. Poster.

Research Support, Awards, and Recognition

- USDA-supported Postdoctoral Associate, University of Florida; waterhyacinth integrated-management project, award 58-6032-2-005 (2026–Present).
- Contributed to research supported by the U.S. Geological Survey Greater Everglades Priority Ecosystem Science Program (2021–2025) and USDA invasive-species and biocontrol funding (2024–2025).
- BGSA DEI Award, University of Florida Department of Biology (2024–2025); Michael L. May Research Grant ($300); Graduate Student Council Travel Grant ($400); Three-Minute Thesis finalist (2022).

Teaching Appointments

Teaching interests: mathematical and computational biology; ecological modeling; quantitative ecology; introductory and non-majors biology; evolution and biodiversity; invasion biology.

University of Florida Summer 2026
Teaching Assistant, BSC 2005 – Biological Sciences for Non-Majors *Gainesville, FL*

- Assessed student projects and scientific writing across genetics, evolution, biodiversity, ecology, conservation, and sustainability.
- Provided rubric-based feedback on evidence, biological reasoning, and communication of scientific ideas to non-specialist audiences.

University of Florida Fall 2025–Spring 2026
Head Teaching Assistant, BSC 2010 – Integrated Principles of Biology I *Gainesville, FL*

- Supported instruction for more than 900 students across two semesters, coordinating grading workflows, office-hour support, and communication within the instructional team.
- Supported consistent course policies and assessment practices; addressed student questions about course content, accommodations, learning platforms, and instructional resources.

University of Florida Spring 2023
Teaching Assistant, BSC 2011L – Integrated Principles of Biology II Laboratory *Gainesville, FL*

- Facilitated laboratory and active-learning activities in organismal biology, plant physiology, and biodiversity.
- Guided students in experimental design, analysis and interpretation of biological data, and communication of laboratory results.

University of Miami Spring 2020
Teaching Assistant, BIL 161 – Evolution and Biodiversity Laboratory *Coral Gables, FL*

- Led laboratory sections focused on evolutionary processes, organismal diversity, and comparative biological observation.
- Evaluated assignments and provided feedback on data interpretation and scientific writing.

Academic Mentoring

Yuexia Lu, M.Sc. Sept. 2026–Present
Academic Mentor *Remote*

- Guide a prospective doctoral applicant in identifying suitable Ph.D. programs, exploring research opportunities, and locating application resources.
- Provide ongoing guidance on application planning and preparation of application materials.

High School Computational Ecology Project 2026–Present
Research Mentor; co-advised with Dr. Donald L. DeAngelis *Remote*

- Guide a student through a NetLogo-based computational ecology project, from defining a tractable research question to designing and interpreting simulation experiments.
- Support the student's skills in model documentation, ecological reasoning, and presentation of results, with the goal of contributing to an ongoing manuscript.

Lijie Yan, M.S. Student, University of Miami Fall 2020–Fall 2021
Peer Research Mentor *Coral Gables, FL*

- Provided individualized guidance on research-focus decisions, graduate coursework, research expectations, and the transition to graduate study.

Teaching Development

Preparing Future Faculty Program Fall 2024
Participant, University of Florida Center for Teaching Excellence *Gainesville, FL*

- Completed training in evidence-based teaching, inclusive pedagogy, course design, and development of an academic teaching portfolio.

Academic and Professional Service

Student support and departmental engagement: Co-created onboarding and peer-support resources for incoming biology graduate students, assisted new students in navigating departmental expectations and resources, and communicated cohort concerns to departmental leadership (2024–2026).

Departmental service: Biology Graduate Student Association Recruitment Committee (2025–2026) and Welcoming Committee (2023–2025); Department of Biology Personnel Support Committee (2022–2026); International Student Workshop (2021–2023).

Journal reviewer: *Biological Invasions*; *Ecography*; *PLOS ONE*; *Scientific Reports*; *Mathematical Biosciences*.

Professional Affiliations and Languages

Ecological Society of America; British Ecological Society; Botanical Society of America; Sigma Xi.

Languages: English and Mandarin.